Chapter 13: MCQ & EDS Preparation

Consolidation for the MCQ Exam and EDS Assignment

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1 Overview

EDS = Experimental Design & Statistics — the assignment due 20 March where you interpret R output and answer questions on design, analysis, and methods.

This chapter consolidates the content you need for two assessments:

1. **MCQ Exam** (Wednesday 11 March): 36 questions — 5 scenario-based investigations (design, IV/DV, hypothesis, test, interpretation) plus 6 general questions.
2. **EDS Assignment** (Due 20 March): Interpret R output and answer questions on hypothesis, design, descriptives, t-test, APA results, data quality, methodological improvements, individual differences, and ecological validity.

The same concepts underpin both: design types, variables, hypotheses, choosing the right test, interpreting p-values and confidence intervals, and clear reporting.

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2 Part 1: MCQ — Design Types and Variables

2.1 Design Types

Design	Description	IV?	Statistical test
Between-participants (independent groups)	Different participants in each condition; random allocation.	Yes — the condition (e.g. warm vs cool room).	Independent-samples t-test
Within-participants (repeated measures)	Same participants in all conditions; order often counterbalanced.	Yes — the condition (e.g. gum vs no gum).	Paired-samples t-test

Design	Description	IV?	Statistical test
Correlational	Two variables measured; no manipulation of an IV.	**No** — there is no independent variable.	Pearson's r (or Spearman's rho for ordinal/non-normal)

Matched pairs: Different people but matched (e.g. on age); sometimes analysed with a paired test if the pairing is meaningful.

2.2 Independent and Dependent Variables

- **IV**: What the researcher manipulates or uses to form groups. Only in experimental designs.
- **DV**: The outcome that is measured.
- In **correlational** designs, do not label one variable as the IV — both are measured variables.

2.3 One-Tailed vs Two-Tailed Hypotheses

- **One-tailed**: Predicts the direction (e.g. "cool room will score higher"; "negative correlation between social media and self-esteem").
- **Two-tailed**: Predicts a difference or relationship without direction (e.g. "there will be a significant difference in test scores between the two temperature conditions").

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3 Part 2: MCQ — Choosing the Test and Interpreting p

3.1 Choosing the Right Test

1. **Correlational design?** (two measured variables, no manipulation) → **Pearson's r** (or Spearman's rho if data are ordinal or not normal).
2. **Same participants in both conditions? → Paired-samples t-test**.
3. **Different participants in each condition? → Independent-samples t-test**.

3.2 p-Values and Significance

- **$p < .05$**: Statistically significant at $\alpha = .05$; reject the null hypothesis; "results support the hypothesis" (if the hypothesis predicted a difference/effect).
- **$p > .05$**: Not statistically significant; do not reject the null.
- **$p < .01$**: Significant; the pattern would occur by chance less than or equal to 1 time in 100.

3.3 Correlation

- **Positive correlation**: Higher on one variable tends to go with higher on the other (e.g. study hours and grades).

- **Negative correlation**: Higher on one tends to go with lower on the other (e.g. social media use and self-esteem).
- **Correlation does not imply causation** — you cannot conclude that X causes Y from a correlation alone (confounds, direction of cause, third variables).

3.4 Counterbalancing

Used in within-participants designs to control for **order effects** (e.g. practice, fatigue) by varying the order of conditions across participants.

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4 Part 3: MCQ — General Concepts

Concept	What you need to know
Operationalisation	Defining a construct by how it is measured (e.g. intelligence = score on a standardised IQ test).
Replication	Another researcher repeating the study; **detailed procedure and materials** help most for replication.
Confounding variable	An uncontrolled variable that co-varies with the IV (e.g. time of day when one group is tested in the morning and the other in the evening).
Null hypothesis	States that there is no significant difference or effect in the population.
$p > .05$	Result is **not** statistically significant at $\alpha = .05$.
Population	The larger group we want to generalise to; the **sample** is the subset we actually test.

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5 Part 4: EDS Section A — Statistical Mechanics

5.1 Stating the Experimental Hypothesis (Q1)

Write a clear hypothesis that refers to both the IV and the DV. It can be directional or non-directional.

- **Good**: “Participants will score higher on the working memory task when their phone is in their bag than when it is visible on the desk.”
- **Good**: “The presence of a smartphone will affect working memory performance.”
- **Avoid**: Stating the null hypothesis or a hypothesis that does not reference both variables.

5.2 Design, IV, DV, Data Type (Q2)

- **Design**: Repeated measures / within-subjects (same participants, two conditions).
- **IV**: Name the variable and **both levels** (e.g. phone visible on desk vs phone in bag).
- **DV**: The measured outcome (e.g. working memory score out of 25).

- **Data type**: Ratio or interval (numeric scale suitable for means and t-tests).

5.3 APA Descriptives and Interpretation (Q3)

- **Report**: "Participants scored higher when their phone was in their bag ($M = 20.30$, $SD = 1.49$) than when their phone was visible ($M = 16.90$, $SD = 4.77$)."
- **Interpret**: Describe the pattern (which condition had higher/lower scores, approximate difference) and note any striking difference in variability (e.g. larger SD in one condition).

5.4 Paired t-Test: Why, Report, Significance, CI (Q4)

- **Why paired?** Same participants completed both conditions; scores are naturally paired.
- **Report**: $t = -2.36$, $df = 9$, $p = .042$ (from output).
- **Significant at $\alpha = .05$?** Yes — $p = .042 < .05$; reject the null.
- **95% CI**: The interval (e.g. [−6.66, −0.14]) estimates the true mean difference. If it excludes zero, the result is consistent with significance. A fully negative interval indicates lower scores in the first condition (e.g. phone visible) than the second (e.g. phone in bag).

5.5 Full APA Results Paragraph (Q5)

Include: name of test, descriptive statistics (M, SD) for each condition, $t(df)$ = value, p = value, 95% CI, and a one-sentence conclusion (e.g. whether the hypothesis was supported).

5.6 Data Quality (Q6)

- **Identify outlier**: Which participant, scores in each condition, and how their difference compares to others (e.g. one participant with a much larger drop in one condition).
- **Plausible reason**: e.g. high distraction, phone dependency, procedural issue.
- **Effect on SD**: An extreme score far from the mean increases the SD of that condition; removing it typically reduces SD.
- **Sensitivity**: Report both with and without the outlier if appropriate; explain that removing an extreme case can sometimes make the test *more* significant (less variability in differences). Justify whether to report one or both analyses and note that the main conclusion may hold either way.

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6 Part 5: EDS Section B — Methods and Theory

6.1 Methodological Improvements (Q7)

Suggest **two** improvements with clear justification. Examples: larger sample size (power, generalisability); controlling for notifications (phones silenced); measuring phone dependency; longer break between conditions; multiple cognitive tasks; screening for engagement.

6.2 Individual Differences (Q8)

Discuss how **heavy vs light phone users** or **younger vs older adults** might differ in the effect, with reasoned theoretical or empirical support (e.g. attentional bias, habituation, age-related cognitive change). Marks are for quality of reasoning, not for a specific prediction.

6.3 Ecological Validity (Q9)

Propose a **more ecologically valid** study: real-world setting and task, with a clear **hypothesis**, **IV** (with levels), **DV**, and **procedure**. Explain why it is more ecologically valid (e.g. real classroom, real learning, natural phone use).

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7 Part 6: Credibility (Brief Context)

Understanding why methods matter helps you see why the MCQ and EDS emphasise correct design, test choice, and reporting:

- **P-hacking**: Using flexibility in analysis to obtain $p < .05$ inflates false positives.
- **HARKing**: Presenting post-hoc ideas as if they were pre-registered predictions undermines credibility.
- **Replication**: Many findings fail to replicate; clear methods and appropriate statistics support replicability.

For more on p-hacking, HARKing, and the replication crisis, see the open science and credibility resources in the module.

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8 Summary Tables

8.1 MCQ: Design → Test

Design	Test
Between-participants (2 groups)	Independent-samples t-test
Within-participants (same people, 2 conditions)	Paired-samples t-test
Correlational (2 measured variables)	Pearson's r

8.2 EDS Section A Checklist

- ☐ Experimental hypothesis (IV + DV)
- ☐ Design type, IV with levels, DV, data type
- ☐ APA descriptives (M, SD) and interpretation
- ☐ Why paired t-test; report t, df, p; significance at $\alpha = .05$; interpret 95% CI
- ☐ Full APA results paragraph
- ☐ Outlier: identify, reason, effect on SD, sensitivity (with/without)

8.3 EDS Section B Checklist

- ☐ Two methodological improvements with justification
- ☐ Individual differences (e.g. heavy/light users, age) — reasoned discussion

- ☐ Ecological validity: hypothesis, IV, DV, procedure, and why more ecologically valid

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9 Further Reading

- Module materials for Weeks 11–16 (design, t-tests, correlation).
- EDS marking scheme (for Section A and B mark allocation and model answers).
- Open science and credibility: pre-registration, replication, effect sizes (see course resources).